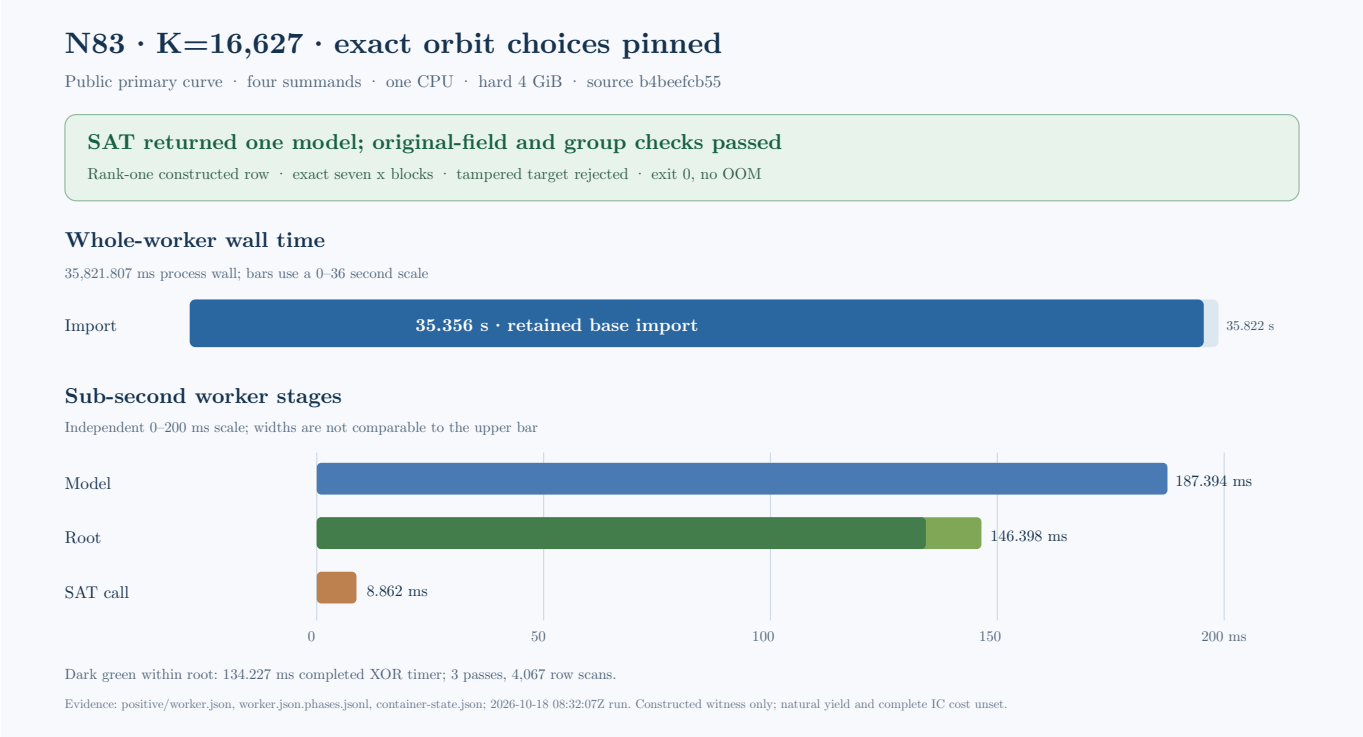
N83 K = 16,627: selector-pinned four-summand control

Run: 2026-10-18 08:32:07–08:32:43 UTC. **Worker result:** `PASS_SELECTOR_PINNED_WITNESS`. This is a constructed relation control on `icv1-f2m83-tm6151469093347-debefd74`. The exact prebuilt signed-Frobenius base has 16,627 columns and 2,760,082 stored points. The accepted source is commit `b4beefcb55f05713d1d63468fcbcd21e4339fa50`; `source-files.sha256`, `inputs-before-run.sha256`, `worker-binary.sha256` and `BUILD_RECEIPT.md` bind the executed worker and public inputs.



Frozen arm and observations

Measure	Recorded value
Resource envelope	one CPU; hard 4 GiB; zero swap; 1,800 s outer cap
Full model before choice pins	201,598 variables; 5,915,238 clauses; 3,901 XOR rows
Base import	35,356.152683 ms
Model construction	187.394292 ms
Root	`Fixpoint` in 146.397875 ms; all 201,598 variables assigned
Root XOR work	3 passes; 134,226,874 ns; 4,067 row scans
SAT call	`Sat` in 8.861792 ms; one model, zero conflicts
Worker wall	35,821.806933 ms
Container	exit 0; no OOM; 2,662,318,080-byte peak

Verification and scope

The worker checked all seven x blocks against the known public points, re-evaluated the original-field S3 chain, lifted the four summands to the stored base, matched the preflight row, obtained rank one and rejected an `R+G` target tamper. The raw worker, phase and progress records are in `positive/`, bound by `positive-files.sha256`. The small direct/observed controls passed; their timing spread is recorded in `VALIDATION.md`. For 20 AB/BA pairs of 100 checked solves per arm, the direct batch median was 8.842667 ms (8.514958–288.700333 ms range) and the observed batch median was 14.477375 ms (8.521667–115.515958 ms range). This shared-host spread is not used as an overhead correction or speed claim.

The separate native `replay/receipt.json` recomputed curve and subgroup membership, seven group x blocks, three original-field S3 equations and the phase-zero relation row from the compressed object. Changed-x and changed-row receipts were rejected. The worker did not export SAT assignment bits, so the separate replay does not check those bits.

The earlier seven-x-block control reached its outer cap while searching. It had a different source revision and no selector or phase pins, so these wall values are not a paired speed ratio. This positive control does not measure natural relation yield, useful natural rank or complete cold IC cost. The next experiment separates selector and phase information under a common source. Canonical end-to-end cost and curve graphs remain unchanged.